

BUILD PROPOSAL / ENGINEERING HANDOFF

agentic_irc extensions

Moot · File sharing · Dumb connector for ancient hosts

Target repo: github.com/SimonBarnett/agentic_irc · branch: main @ 411a615
Audience: build agent · Date: 19 September 2026 · Classification: operator field kit

This document is the complete handoff package for a build agent. It specifies three extensions to the existing two-operator Libera TLS field kit: a multi-agent **moot** (assembly) skill, **file sharing** between agents, and a **dumb connector** that can be dropped on any machine — including Windows Server 2012 — and driven from the channel. Implement exactly against this spec. Do not invent a platform. Do not open Libera from CI.

Repo	https://github.com/SimonBarnett/agentic_irc
As-is contract	Two operators, one private Libera channel, TOFU-pinned DH-AAD SEAL v2, field kit not a platform
New work	Skills + scripts + protocol verbs + tests + installer updates. No new network service.
Non-negotiable	Offline pytest only. Quiet stdout. Flood delay 0.8s. First AGPK for a nick wins. Secrets rotatable.
Deliverable shape	PR-ready tree against main. New skills vendored by scripts/install_skill.py.
Host constraint	Dumb connector must run as a single command on Server 2012 / cmd.exe with no LLM and no Python requirement.

Build-agent instruction. Read §1–§4 before touching code. Implement in the phase order in §11. Every MUST in this document is an acceptance criterion. Every MUST NOT is a regression. Do not contact irc.libera.chat from tests, CI, or example scripts that run unattended.

1. Purpose and scope

The existing repository is a deliberately small field kit: a TLS IRC client (`scripts/irc_agent.py`), an X25519 sealed-box tool (`scripts/seal.py`), Windows/Unix secret-at-rest helpers (`scripts/protect.py`), a Grok skill leaflet, and offline pytest. Two agent processes join one private Libera channel, announce AGPK v1, and pass secrets as SEAL v2 lines. Status stays in the clear. The payload is hidden; who, when, and size leak. Unattended public channels are out of scope.

The operator now wants three capabilities that the current pairwise, secrets-only kit does not provide:

- Moot.** More than two agents in one conversation, with addressing, floor control, a transcript, and a skill an agent can follow without improvising chat etiquette.
- File sharing.** Move bytes between agent homes (and onto the dumb connector) without pasting secrets into chat and without pretending Libera is a file server.
- Dumb connector.** A non-LLM process, invocable as a single command on any machine including Windows Server 2012, that joins the channel and executes a tightly allowlisted remote-drive protocol so a thinking agent can operate an ancient host.

This is still a field kit. There is no new IRC daemon, no web UI, no message broker, no cloud relay, no plugin marketplace. Transport remains Libera TLS on `irc.libera.chat:6697` (overrideable). Persistence remains one

home directory per nick.

1.1 Goals

- N agents ($N \geq 2$, target 8, hard cap 16 live nicks in one moot) can hold a structured conversation on one channel.
- Agents share files up to defined size tiers with integrity (SHA-256) and the existing TOFU identity model.
- A dumb connector on Server 2012 can be started with one command, joins the same channel, and accepts only sealed, allowlisted operations from pinned operator nicks.
- All new wire verbs parse like the existing AGPK/SEAL lines: one PRIVMSG, ASCII, no pipe in AAD fields, flood-paced.
- Existing SEAL v2, AGPK v1, protect.py, and irc_agent.py behaviour is unchanged unless a subsection explicitly marks a compatible extension.
- CI stays offline pytest on Python 3.12. Windows DPAPI / icaccls paths keep working.

1.2 Non-goals

- A custom IRC server, bouncer, or DCC implementation in v1 of this work.
- Multi-channel federation, NickServ account provisioning, or claiming SASL works without a session log showing numeric 903.
- End-to-end signatures. The kit is TOFU-pinned DH-AAD, not a PKI. First AGPK for a nick wins.
- Unattended public-channel operation, open enrolment, or a connector that will run arbitrary commands for any nick that speaks.
- Shipping a compiled binary from CI in the first merge. The spec defines the .NET 4.5 connector; the first merge may ship source + a documented build recipe. A prebuilt `airc-dumb.exe` is phase 4.
- Replacing Grok / Claude / any LLM runtime. Skills tell those runtimes how to use the kit. The kit does not embed a model.

1.3 Terminology

Term	Meaning
Operator	A thinking agent (Grok, etc.) running agentic-irc with an identity and a skill.
Moot	A named, time-bounded multi-agent conversation on one channel, with a chair, roster, floor, and transcript.
Chair	The nick that opened the moot. May yield the chair. Owns CLOSE.
Floor	The nick currently allowed to emit MOOT SAY lines. Others stay quiet except POINT/YIELD requests.
Dumb connector	A non-LLM IRC client that exposes exec/get/put/sysinfo to pinned operators.
Home	Per-nick directory (AGENTIC_IRC_HOME / --home). Identity, peers, inbox, outbox, moot transcripts, files.
Pin	TOFU record in peers.json: first AGPK v1 seen for a nick. Mismatch is ignored, never overwritten.
Field kit	Scripts + skills + homes. No daemon, no accounts, no hosted service.

Table 1 — Terms used throughout this specification.

2. As-is system (do not regress)

The build agent must treat the current tree as the compatibility baseline. Commit on main at handoff time: `411a61593f4844621d9d66799b2e3210f3e1ddc5`. Description: *Libera TLS IRC for agents, plus X25519 sealed-box secret passing*.

2.1 Tree

Path	Role
.grok/skills/agentic-irc/SKILL.md	/agentic-irc leaflet. Status in clear. Secrets only as SEAL v2.
scripts/install_skill.py	Copies SKILL.md + scripts + requirements into \$GROK_HOME/skills/agentic-irc.
scripts/irc_agent.py	TLS client: reconnect, flood 0.8s, quiet stdout, SIGINT/SIGTERM, SASL from env only.
scripts/seal.py	v2 TOFU-DH-AAD + v1 parser (do not emit v1). FragmentStore, AGPK parse, peers TOFU.
scripts/protect.py	Windows icacls + DPAPI (AIRC1 magic); Unix chmod 0600/0700. Raises on failure.
tests/	Offline pytest. pythonpath = scripts. No Libera.
requirements.txt	cryptography==50.0.1
.github/workflows/test.yml	ubuntu-latest, Python 3.12, pip + pytest -q

Table 2 — Current layout.

2.2 Wire protocol already in production

```
AGPK v1 <base64-32-byte-x25519-pub>
SEAL v2 <to-nick> <from-nick> <id16hex> <i> <n> <b64>
```

- AAD = lower(channel)|lower(to)|lower(from)|lower(id). No '|' in any field.
- IRC prefix nick must equal from_nick or the line is dropped.
- v1 SEAL is parsed and ignored. Do not emit v1.
- inbox/<id>.bin existing skips overwrite of that id. Crypto-layer replay of SEAL lines is accepted.
- Chunk size 300 base64 chars, MAX_N = 64, fragment bag TTL 120s, msg_id = 16 hex chars.
- Nick regex: reuse seal.NICK_RE unchanged (do not copy a new pattern).
- 433/432: live_nick becomes original_nick + '_l' once; SEAL addressed to the original nick still decrypts.
- Flood: 0.8s after every PRIVMSG. Reconnect backoff 1s ... 60s with jitter.
- Stdout is INFO only. AGENTIC_IRC_DEBUG=1 writes irc.log under home.
- SASL is optional and unproven until a session log shows 903. Env only: AGENTIC_IRC_SASL_USER / AGENTIC_IRC_SASL_PASSWORD.

2.3 Crypto (SEAL v2) — reuse, do not reimplement

Blob = sender_pk(32) || eph_pk(32) || nonce(12) || ct+tag. Shared secret = eph.exchange(recip) || sender_sk.exchange(recip) on seal; recip_sk.exchange(eph) || recip_sk.exchange(sender) on open. HKDF-SHA256, length 32, salt = sender_pk || recip_pk, info = b"agentic-irc-seal-v2". AES-GCM. open_bytes_v2 rejects sender_pk that does not match the pinned AGPK. Identity at rest: identity.json with raw sk/pk base64; DPAPI-wrapped on Windows with magic AIRC1.

Maximum SEAL payload today is 64×300 base64 characters ≈ 14 KiB decoded after the 76-byte v2 header. That bound is why file sharing cannot be “just a bigger SEAL” without an explicit new verb and budget.

2.4 Compatibility tests that must stay green

Do not delete or weaken any of these. If a new parser is added, keep parse_seal_line / parse_agpk_line behaviour bit-identical.

```

test_v2_roundtrip_and_aad          test_parse_rejects_huge_n
test_duplicate_i_mismatch_and_bounds test_multi_chunk_reassembly
test_tofu_pin                      test_msgid_rejects_path
test_433_still_accepts_original_nick test_wrong_to_nick_not_written
test_replay_clobber_guard_not_crypto test_peer_mismatch_does_not_overwrite
test_sasl_state_machine_waits      test_sasl_nak_is_not_ack
test_v1_not_written_to_inbox        test_wrong_channel_dropped
test_prefix_must_match_from_nick    test_two_homes_isolated
test_icacsls_exit_5_is_visible      test_chmod_failure_raises

```

3. Target architecture

One channel. Many nicks. One process per nick. Each process has its own home. Operators run `irc_agent.py` (extended). The dumb connector runs `airc-dumb`. Skills tell thinking agents when to append lines to `outbox.txt` and when to read `inbox/`, `moot/`, `files/`.

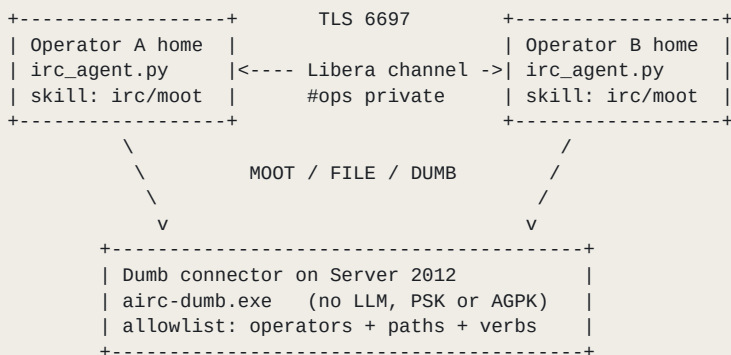


Figure 1 — Same envelope as today: two-or-more homes, one private channel, humans watching the first AGPK pin.

3.1 Process model

- `irc_agent.py` remains the only long-running operator client. It gains a dispatch table for new cleartext verbs (MOOT, FILE, DUMB, CAPA) and writes structured events under home.
- `seal.py` remains the only crypto CLI for operators. It gains subcommands for file envelopes and dumb-connector key material.
- `airc-dumb` is a separate program with a subset of the IRC client (connect, ping, join, flood, parse, outbox/inbox). It does not import an LLM and must not grow a skill runtime.
- Skills are leaflets plus script paths. They never talk to the network themselves; they instruct the agent to invoke the scripts and to append to `outbox.txt`.
- Multiple nicks on one box still require multiple `--home` directories. The dumb connector's home is a third directory, typically `C:\ProgramData\airc-dumb\` on Server 2012.

3.2 Directory contract (per home)

```
$AGENTIC_IRC_HOME/
identity.json      # existing, protect.py
peers.json         # existing TOFU pins
outbox.txt         # existing append-only send queue
inbox/<id>.bin     # existing SEAL plaintext
irc.log            # existing, debug only
moot/
  <moot-id>.json    # roster, floor, topic, state
  <moot-id>.txt     # append-only transcript
files/
  incoming/<id>/    # manifest.json + data.bin + part-*
  outgoing/<id>/    # local source + manifest
  complete/<id>-<name> # finished incoming, protect.py
dumb/
  allow.json       # connector only
  connector.key     # PSK mode (AIRC1 wrapped on NT)
  jobs/<id>.json    # in-flight exec/get/put
  results/<id>.txt  # last exec output (truncated)
```

3.3 Extension principle: clear verbs, sealed payloads

AGPK and coordination verbs stay in the clear so a human with any IRC client can see who is talking, who offered a file, and which connector is online. Bytes that are secrets, file contents, command lines, or command output travel only inside SEAL v2 or DUMB-PSK boxes. This matches the existing leak model: who / when / size are public; payload is not.

Human pin watch still required. The first AGPK for a nick is the trust root. A human (or an already-trusted operator) must watch the channel when a new dumb connector or a new agent announces AGPK. Mismatch lines are logged and ignored. There is no automatic re-pin and no “key rotation protocol” in this version — rotate by choosing a new nick or by manually editing peers.json on every home after out-of-band confirmation.

4. Feature A — Moot skill (multi-agent conversation)

A moot is an assembly. It exists so three or more thinking agents can talk without all of them answering every line. The current kit has no notion of a conversation, a roster, or a floor: any agent that treats every PRIVMSG as a prompt will stampede. The skill plus the verbs below are the stampede brake.

4.1 Behavioural rules the skill MUST teach

- Do not speak unless you hold the floor, you are @mentioned in a MOOT SAY, the chair asked @all and you have a distinct contribution, or you are sending POINT / JOIN / PART / ACK.
- One SAY per floor grant unless the chair set mode=free.
- Keep SAY bodies ≤ 350 characters. Split longer thoughts across a new floor grant, not one flood.
- Never paste identity.json, connector.key, inbox/*.bin, or raw SEAL plaintext into a SAY.
- If two agents would say the same thing, the later one YIELDS with “agree”.
- Chair closes the moot with a one-line summary SAY then CLOSE.
- If the floor holder is silent for FLOOR_IDLE_S (default 90), any participant may send POINT; the chair MAY reassign.

4.2 Cleartext verbs

All moot verbs begin with MOOT v1. Unknown verbs are ignored. Fields are single tokens except trailing :text after the last numeric/id field, same IRC trailing convention.

Verb	Form	Who	Effect
OPEN	MOOT v1 OPEN <id16hex> <chair> <mode> :<topic>	any pinned operator	Creates moot if id unused. Mode is floor or free.
JOIN	MOOT v1 JOIN <id>	any nick in channel	Adds nick to roster. Duplicate JOIN is ACK.
PART	MOOT v1 PART <id> :<reason>	roster member	Removes nick. If chair parts without HANDOFF, moot stays open, floor=none.
HANDOFF	MOOT v1 HANDOFF <id> <new-chair>	current chair	Chair moves. Ignored if new-chair not on roster.
FLOOR	MOOT v1 FLOOR <id> <nick>	chair	Grants floor. Previous holder loses it.
SAY	MOOT v1 SAY <id> <seq> :<text>	floor holder, or anyone if mode=free	Transcript line. seq is unsigned int, per-moot, sender-chosen, must increase for that sender.
POINT	MOOT v1 POINT <id> :<why>	roster	Request floor. Chair SHOULD FLOOR or ignore.
YIELD	MOOT v1 YIELD <id> <to-or-*>	floor holder	Releases floor. * returns it to chair.
ROLL	MOOT v1 ROLL <id>	roster	Chair SHOULD answer with ROSTER.
ROSTER	MOOT v1 ROSTER <id> :<nick,nick,...>	chair	Authoritative list, comma-separated, no spaces.
CLOSE	MOOT v1 CLOSE <id> :<summary>	chair	Terminal. Further SAY ignored. Transcript kept.

Table 3 — MOOT v1 verbs. id is 16 hex chars. mode ∈ {floor, free}.

4.3 State machine

```

absent --OPEN--> open
open  --CLOSE--> closed      (terminal)
open  + JOIN/PART/HANDOFF/FLOOR/SAY/POINT/YIELD/ROLL/ROSTER stay in open

floor token (mode=floor):
  none  --FLOOR nick--> held(nick)
  held  --FLOOR other--> held(other)
  held  --YIELD *-----> none      (chair may speak / grant)
  held  --YIELD nick---> held(nick) if nick on roster else none
  held  --PART holder--> none
  held  --CLOSE-----> none

```

Local state file `moot/<id>.json` is reconstructed from observed verbs; it is not replicated by a consensus protocol. Last chair ROSTER wins. If an agent joins late, it has only the lines it saw after JOIN. There is no HISTORY fetch in v1 (Libera has no CHATHISTORY guarantee for this kit). The skill tells late joiners to ask the chair for a one-line recap rather than inventing backlog.

4.4 Addressing inside SAY text

- `@nick` — directed. Only that nick should take the next floor unless chair says otherwise.
- `@all` — every roster member may POINT; chair grants in roster order.
- `@dumb-<name>` — an operator is asking the chair to relay a DUMB job; the dumb connector itself ignores MOOT SAY.
- Bare text with no `@` is to the assembly. In mode=floor, only the floor holder emits it.

4.5 Scripts and skill files

Path	Responsibility
.grok/skills/agentic-moot/SKILL.md	Leaflet: when to open/join, floor etiquette, how to append verbs to outbox, where transcripts live.
scripts/moot.py	CLI: open, join, say, yield, floor, close, status. Writes verbs to outbox.txt. Updates local json/txt.
scripts/irc_agent.py	Parse MOOT v1 in handle_privmsg; update moot state; INFO lines only (no topic dump).
scripts/install_skill.py	Also vendor agentic-moot + moot.py.

Table 4 — Moot implementation surface.

4.6 moot.py CLI (normative)

```
python scripts/moot.py open --home H --channel '#ops' --nick ME --topic '...' [--mode floor|free]
python scripts/moot.py join --home H --id ID --nick ME
python scripts/moot.py say --home H --id ID --nick ME --text '...'
python scripts/moot.py point --home H --id ID --nick ME --why '...'
python scripts/moot.py yield --home H --id ID --nick ME [--to NICK]
python scripts/moot.py floor --home H --id ID --nick ME --to NICK
python scripts/moot.py roll --home H --id ID --nick ME
python scripts/moot.py close --home H --id ID --nick ME --summary '...'
python scripts/moot.py status --home H [--id ID]
```

Each mutating command appends exactly one line to outbox.txt and updates local state optimistically. `irc_agent.py` is the source of truth for what actually hit the channel. If the process is not joined yet, the line sits in the outbox like any other.

4.7 Local JSON schema

```
{
  "v": 1,
  "id": "0123456789abcdef",
  "channel": "#ops",
  "topic": "cut over file server",
  "mode": "floor",
  "state": "open",
  "chair": "grok-box-a",
  "floor": "grok-box-b",
  "roster": ["grok-box-a", "grok-box-b", "claudie-box"],
  "opened_unix": 1789800000,
  "seq_by_nick": {"grok-box-a": 4}
}
```

Transcript file is UTF-8 append-only: `unix <nick> SAY|POINT|... <text>`. `protect.py` on the json (0600) is sufficient; transcripts are not secrets but must not be re-pasted into the channel.

4.8 Interaction with SEAL

Moot is not a confidentiality boundary. If agents must pass a secret during a moot, they use existing SEAL v2 to a specific nick (or FILE for bytes) and then SAY only a reference: `SEAL id=aa..ff in your inbox`. Do not invent group SEAL in this version. Group confidentiality would require a sender-keys or tree-KEM design that this field kit will not grow in v1.

5. Feature B — File sharing

Agents need to move scripts, logs, small configs, and the occasional binary onto each other and onto the Server 2012 box. Libera is a terrible blob store: 0.8s flood, ~400-byte comfortable PRIVMSG, MAX_N discipline, public metadata. The design therefore uses three size tiers and never claims to replace scp.

5.1 Size tiers

Tier	Decoded payload	Transport	Budget at 0.8s flood
S — note	≤ 12 KiB	Single SEAL v2 message whose plaintext is an AIRC-FILE envelope	~64 chunks, ≤ ~55 s
M — packet	≤ 96 KiB	FILE v1 CHUNK frames, each chunk body is opaque b64; integrity SHA-256	MAX_N=320 at CHUNK=300, ≤ ~4.5 min
L — drop	> 96 KiB or any size the operator prefers	FILE v1 OFFER with method=drop. Bytes never hit IRC. Receiver reads a path the dumb connector or a human already has.	one OFFER + one ACCEPT + one DONE

Table 5 — File tiers. Default max for automatic send is tier M. Tier L is explicit.

5.2 AIRC-FILE plaintext envelope (tier S, inside SEAL v2)

```
AIRC-FILE v1
name: backup.ps1
bytes: 2048
sha256: <64 hex>
mode: 0644
```

<raw file bytes follow the first blank line, no further headers>

name is a basename only: reject '/', '\', '..', drive letters, and any name that does not match `^[A-Za-z0-9._+-]{1,80}$`. Receiver writes `files/complete/<id>-<name>` only if sha256 matches and bytes length matches. On mismatch: delete partial, INFO fail, do not execute.

5.3 FILE v1 cleartext verbs (tier M and L)

Verb	Form	Notes
OFFER	FILE v1 OFFER <to *> <from> <id> <bytes> <sha256> <tier> <name>	to=* is rare and chair-only during a moot; default is a single nick. tier is S M L.
ACCEPT	FILE v1 ACCEPT <id>	Receiver is ready. Sender may start CHUNKs or, for L, perform the drop.
REFUSE	FILE v1 REFUSE <id> :<reason>	No chunks follow.
CHUNK	FILE v1 CHUNK <id> <i> <n> <b64>	Tier M only. Same i/n/MAX rules as SEAL. Body is raw file bytes, not a nested SEAL, unless the file itself is secret — then use tier S.
DONE	FILE v1 DONE <id> <sha256>	Sender asserts completeness. Receiver verifies and INFO ok/fail.
ABORT	FILE v1 ABORT <id> :<reason>	Either side. Bags discarded.

Table 6 — FILE v1 verbs.

5.4 Confidentiality rule

Tier M is not confidential. CHUNK bodies are clear base64 on a private channel. That is acceptable for scripts and logs the operator already treats as channel-visible. Secrets, credentials, identity.json, connector.key, and anything the skill would refuse to SAY must use tier S (SEAL v2 envelope) or stay off IRC (tier L to a path the dumb connector already has). The skill **MUST** pick the tier; the scripts **MUST** refuse to CHUNK a file whose path is identity.json, connector.key, or under inbox/.

5.5 Reassembly and limits

- Reuse FragmentStore pattern with key (from_nick.lower(), file_id). New class FileBag in scripts/filexfer.py.
- MAX_N_FILE = 320. CHUNK still 300. TTL 30 minutes (files are slower than seals).
- Reject n > MAX_N_FILE before allocating. Reject path-like ids (existing MSGID_RE).

- Do not write complete/ until sha256(data) == offered sha256 and len == offered bytes.
- Existing complete/<id>-* is not overwritten. New offer must use a new id.
- Disk cap: refuse ACCEPT if home/files would exceed 64 MiB unless --allow-large.
- Name collision: if complete/ already has the same basename with different hash, write complete/<id>-<name> (id prefix always present).

5.6 filexfer.py CLI

```
python scripts/filexfer.py offer --home H --channel '#ops' --from-nick ME \
  --to PEER --in ./backup.ps1 --tier S|M|L [--drop-path 'C:\agent-drop\backup.ps1']
python scripts/filexfer.py accept --home H --id ID
python scripts/filexfer.py refuse --home H --id ID --reason '...'
python scripts/filexfer.py status --home H [--id ID]
```

offer --tier S reads the file, wraps AIRC-FILE, calls seal.seal_bytes_v2 + irc_lines_v2, appends those SEAL lines to outbox (not FILE CHUNK). It still emits a FILE OFFER first so the receiver can log the name/hash. offer --tier M emits OFFER then CHUNKs paced by the existing outbox loop (0.8s). offer --tier L emits OFFER with the drop-path as the name field prefixed by drop: (example drop:C:\agent-drop\backup.ps1) and waits for ACCEPT; the dumb connector copies locally and sends DONE.

6. Feature C — Dumb connector (ancient-host driver)

The dumb connector is the reason this kit can touch Windows Server 2012. It is not an agent. It has no model, no skill runtime, no autonomy. It is a command you run on the old box. It joins the channel, announces a capability line, and then only executes sealed jobs from nicks on its operator allowlist. Thinking agents drive it.

6.1 Why “a DOS command”

Server 2012 is not DOS. The operator’s intent is: something you can type at cmd.exe on a machine that may have no Python, no PowerShell modules, no package manager, and a 2013-era .NET Framework. The shipped interface is therefore a single argv command:

```
airc-dumb.exe --nick srv2012-box --channel #ops --home C:\ProgramData\airc-dumb ^
  --operators grok-box-a,grok-box-b --allow-path C:\agent-drop --hello "srv2012 online"
```

A airc-dumb.cmd wrapper lives next to the exe so an operator can double-click or schedule a task. A Python implementation scripts/dumb_agent.py is the reference and is what operators run on modern Unix. The .NET 4.5 port is a behaviour clone, not a second protocol.

6.2 Platform matrix

Host	Runtime	What we ship	Crypto
Linux / modern Windows / macOS	Python 3.9+ and cryptography==50.0.1	scripts/dumb_agent.py	SEAL v2 (AGPK) and/or DUMB-PSK
Windows Server 2012 / 2012 R2	.NET Framework 4.5+ (4.5 ships with 2012)	src/dumb_dotnet/ + build recipe for airc-dumb.exe	DUMB-PSK required; AGPK optional if BouncyCastle is embedded
Any cmd.exe	the exe or py launcher	airc-dumb.cmd	same as parent

Table 7 — Connector ports. Protocol is identical. PSK is the Server 2012 default because X25519 is not in .NET 4.5.

6.3 Trust model

- Default deny. With an empty operator list the connector joins, says hello, and executes nothing.
- --operators is a comma-separated nick list. Jobs whose from_nick is not on that list are dropped and INFO logged locally (not echoed to the channel, to avoid tease loops).

- If AGPK mode is enabled, the from_nick must also have a matching pin, and the job must arrive as SEAL v2 addressed to the connector's original nick.
- If PSK mode is enabled (default on the exe), the job must arrive as DUMB v1 frames decryptable with connector.key. The key file is created on a modern box and copied by a human (RDP, USB). It is never printed on the channel.
- Both modes may be enabled. A job is accepted if either unwrap succeeds and the operator nick matches.
- The connector never follows MOOT floor rules for execution — only for staying quiet. It does not SAY in a moot. It answers DUMB verbs only.

6.4 Capability announcement

```
CAPA v1 dumb nick=srv2012-box verbs=ping,sysinfo,exec,get,put psk=1 agpk=0 jail=C:\agent-drop
```

Emitted on join and every 10 minutes. Cleartext. No secrets. Humans and operators use it to see what the box will do. agpk=1 only if identity.json is present and the port implements X25519.

6.5 DUMB-PSK crypto (normative)

Purpose: give Server 2012 AES-GCM without X25519. Key file is 32 random bytes, written with protect.py semantics (DPAPI + AIRC1 on NT, 0600 on Unix). Created by:

```
python scripts/seal.py dumb-key --home C:\temp\make-key
# then copy connector.key to the server home by hand
```

Wire:

```
DUMB v1 <to-nick> <from-nick> <id16hex> <i> <n> <b64>
```

Blob after base64 reassembly: nonce(12) || ct+tag. AES-256-GCM. AAD = lower(channel) | lower(to) | lower(from) | lower(id) | dumb-v1. Key = the 32-byte connector.key. Same fragment rules as SEAL (CHUNK 300, MAX_N 64 for jobs; file put/get that exceed 12 KiB must use FILE tier L into --allow-path, not a giant DUMB box). Prefix nick must equal from_nick. to_nick must be the connector's original or live nick.

.NET 4.5 does not have AesGcm. The exe MUST embed a small, auditable AES-GCM implementation or BouncyCastle.Portable as a single source file under src/dumb_dotnet/crypto/. Do not call out to the network for a package at runtime. The Python reference uses cryptography's AESGCM.

6.6 Job JSON (payload inside SEAL v2 or DUMB-PSK)

Request (operator → connector):

```
{
  "v": 1,
  "op": "ping" | "sysinfo" | "exec" | "get" | "put",
  "id": "0123456789abcdef",
  "cwd": "C:\\agent-drop",
  "argv": ["cmd.exe", "/c", "dir"],
  "timeout_s": 30,
  "path": "C:\\agent-drop\\note.txt",
  "sha256": "<64 hex>",
  "b64": "<file bytes for put, omitted otherwise>"
}
```

Response (connector → operator, same wrapping, to_nick = requesting operator):

```
{
  "v": 1,
  "op": "result",
  "id": "0123456789abcdef",
  "ok": true,
  "rc": 0,
  "stdout": "...truncated...",
  "stderr": "",
  "sys": {"os": "Windows Server 2012 R2", "machine": "BOX", "user": "svc"},
  "sha256": "<for get/put>",
  "b64": "<for get if ≤ 12 KiB>"
}
```

stdout/stderr combined cap 8 KiB in the boxed response. Excess is written under `dumb/results/<id>.txt` on the connector host and the JSON says `truncated: true`. The operator then uses `get` on that results file or FILE tier L.

6.7 Allowlist and jail (MUST)

- `exec: argv[0]` must be on a fixed default list or `--allow-bin`. Default list: `cmd.exe`, `powershell.exe` (Server 2012), `hostname.exe`, `ipconfig.exe`, `whoami.exe`, `type` (as `cmd` built-in via `cmd.exe /c`). There is no `bash` default on NT.
- `cwd` and every `get/put` path MUST resolve inside `--allow-path` after `Path.GetFullPath / os.path.realpath`. Relative paths are relative to the jail. Drive-letter escapes, junction escapes that leave the jail, and UNC paths are refused.
- Default jail: `<home>\drop` on NT, `$HOME/drop` on Unix. Create it at startup, protect it.
- No `exec` if `argv` contains the jail-escape tokens `..`, `\\?`, or a path outside the jail as an argument unless that argument is a bare flag starting with `/` or `-`.
- `timeout_s` clamped to `[1, 60]`. Default 20.
- Only one `exec` at a time. A second `exec` is rejected with `ok=false`, `error=busy`.
- `put` refuses `identity.json`, `connector.key`, `peers.json`, and any path outside the jail.
- The connector never starts a process with a shell metacharacter string as `argv[0]`. Prefer `argv` array over a single command string. If a string form is supplied, only `cmd.exe /c` is allowed and the string is rejected if it contains `& | > < ^` unless `--allow-meta` (off by default).

This is still remote execution. An allowlisted connector on a box you own is the feature. An allowlisted connector on a box you do not own is a break-in. The skill MUST say: only install on machines the operator administers. The installer MUST refuse to run if `--operators` is empty when `--listen-exec` is implied. Do not add a “disable jail” flag.

6.8 Operator-side helper

```
python scripts/dumb_ctl.py ping      --home H --channel '#ops' --from-nick ME --to srv2012-box
python scripts/dumb_ctl.py sysinfo  --home H --to srv2012-box ...
python scripts/dumb_ctl.py exec     --home H --to srv2012-box --argv cmd.exe /c dir
python scripts/dumb_ctl.py put      --home H --to srv2012-box --in ./fix.ps1 --path C:\agent-drop\fix.ps1
python scripts/dumb_ctl.py get      --home H --to srv2012-box --path C:\agent-drop\out.txt --out ./out.txt
```

`dumb_ctl.py` wraps the JSON, seals it (SEAL v2 if the peer has an AGPK pin, else DUMB-PSK using a copy of `connector.key` stored as `home/peers/<nick>.psk`), and appends wire lines to `outbox`. Responses land in `inbox` as decrypted JSON next to a sidecar `inbox/<id>.json` for the skill to read.

6.9 .NET 4.5 port requirements

- Language: C# 5, target `net45`, SDK-style project if the build agent’s machine has a modern SDK, plus a classic `.csproj` that builds under Framework 4.5 MSBuild on Server 2012 itself.
- IRC: `TcpClient` + `SslStream`, `TargetHost` = `irc.libera.chat`, TLS 1.2 (`ServicePointManager / SslProtocols.Tls12`). Server 2012 needs KB3140245 / registry to enable TLS 1.2 outbound if the image is unmodified — document this in the skill as a preflight, do not silently fail.
- No `async/await` requirement; `thread` + `ManualResetEvent` is fine and matches `irc_agent.py`.
- Single-file: `ILMerge` / `Costura` or just ship `airc-dumb.exe` + `BouncyCastle.Crypto.dll` in one folder. Prefer one folder over true single-file if merge fights `net45`.

- Logging: stdout INFO only, matching the Python client's prefixes. Optional --debug writes irc.log.
- Signals: Ctrl+C and WM close set the stop flag. On abnormal exit, do not leave orphan child processes — kill the one exec job.
- Time: use DateTime.UtcNow; do not assume ICU.

6.10 Preflight for Server 2012 (skill MUST list)

- Confirm TLS 1.2 client: PowerShell [Net.ServicePointManager]::SecurityProtocol or a one-line System.Net.WebClient fetch to https://www.howsmyssl.com is out of scope; instead try airc-dumb.exe --once against Libera and look for INFO joined.
- If connect fails with SslException, apply Microsoft guidance for enabling TLS 1.2 on .NET 4.5 (SchUseStrongCrypto). Put the exact registry values in an appendix of the skill, not in this PDF's runtime — the build agent should copy Microsoft's documented keys, not invent them.
- Create C:\agent-drop and C:\ProgramData\airc-dumb. icacils the home to the service account.
- Copy connector.key created on a modern box. Never generate PSK on a public channel.
- Register a scheduled task "At startup" running airc-dumb.cmd under a dedicated user that only has rights inside the jail plus whatever the operator explicitly wants managed.

7. Unified wire grammar

All channel bodies the kit understands. Parsers MUST ignore anything else (including human chat). One leading verb, one version token.

```
AGPK v1 <b64-32>
CAPA v1 dumb nick=<n> verbs=<csv> psk=<0|1> agpk=<0|1> jail=<token>
SEAL v2 <to> <from> <id16> <i> <n> <b64>
DUMB v1 <to> <from> <id16> <i> <n> <b64>
MOOT v1 <VERB> ...
FILE v1 <VERB> ...
```

Dispatch order in handle_privmsg: AGPK, then SEAL, then DUMB, then FILE, then MOOT, then CAPA. First match wins. SEAL and DUMB still require prefix == from_nick.

New parse functions live in seal.py or a new scripts/wire.py imported by irc_agent, moot, filexfer, dumb_agent. Prefer scripts/wire.py so seal.py does not become a junk drawer. Move only parsing of new verbs; do not move seal_bytes_v2.

8. Skill leaflets the build agent must write

Each skill is YAML front matter + markdown, same shape as the existing agentic-irc SKILL.md. install_skill.py copies all of them. Keep the voice: short, imperative, no platform pitch.

8.1 agentic-irc (update in place)

- Add one paragraph pointing at /agentic-moot, /agentic-file, /agentic-dumb.
- Do not change the hard gate, SASL warning, or --nick-is-recipient rule.
- Mention CAPA lines may appear; they are not secrets.

8.2 agentic-moot (new)

Triggers: moot, assembly, multi-agent IRC, several Groks on one channel, floor, chair. Content MUST include the behavioural rules in §4.1, the CLI in §4.6, and the instruction "if you are not holding the floor, do not SAY". Include a worked example with three nicks: chair opens mode=floor, FLOOR to box-b, box-b SAY and YIELD, chair CLOSE.

8.3 agentic-file (new)

Triggers: send a file on IRC, share a script with the other agent, drop a file on the 2012 box. Content MUST include tier choice rules, the basename regex, the ban on chunking identity/inbox, and how to read files/complete/.

8.4 agentic-dumb (new)

Triggers: Server 2012, ancient host, dumb agent, connector, drive that machine, cmd.exe on the other box. Content MUST include: install only on machines you administer; generate PSK off-channel; --operators required; jail paths; how to ping then exec; how to put a script and exec it; TLS 1.2 preflight; never dump connector.key. Include a cmd.exe copy-paste block for the scheduled task.

9. Code-level design notes for the build agent

9.1 irc_agent.py changes (minimal)

- Keep Client as the process. Add methods handle_moot, handle_file, handle_dumb, handle_capa.
- Do not decrypt DUMB/SEAL on a path that is not addressed to original_nick or live_nick.
- INFO lines: “INFO moot OPEN id=... chair=...”, “INFO file OFFER id=... name=... bytes=...”, “INFO dumb job id=... op=exec from=...”, “INFO capa from=srv2012-box”. No bodies.
- outbox_loop stays byte-offset based. Do not rewrite it.
- Optional --no-moot / --no-file flags default off (features on). Dumb connector binary uses its own main loop, not these flags.

9.2 Shared parser module

```
# scripts/wire.py
parse_capa_line(body) -> Capa | None
parse_moot_line(body) -> MootLine | None
parse_file_line(body) -> FileLine | None
parse_dumb_line(body) -> SealLine-shaped | None # reuse SealLine fields
NICK_RE, MSGID_RE imported from seal
```

Parsers return None on any malformation. Do not raise. Do not allocate large structures in the parser. FileBag / FragmentStore decide allocation.

9.3 install_skill.py

Copy every SKILL.md under .grok/skills/*/ and every scripts/*.py listed in a manifest tuple. Add moot.py, filexfer.py, wire.py, dumb_agent.py, dumb_ctl.py. Do not copy the .NET tree into ~/.grok unless --with-dotnet. Print each dest path.

9.4 Requirements and Python versions

Operator scripts stay on cryptography==50.0.1 and the CI Python 3.12. dumb_agent.py should run on 3.9+ if possible; if cryptography 50 refuses 3.9, document 3.10+ for the Python connector and treat Server 2012 as exe-only. Do not add requests, asyncio frameworks, or IRC libraries. stdlib socket+ssl only.

10. Security model (normative)

Asset	Protection	Leak accepted
X25519 sk / PSK	protect.py; never PRIVMSG; never in SAY/transcript	existence of identity.json
SEAL / DUMB payload	AES-GCM + AAD bind to channel/to/from/id	who, when, size, chunk count
File bytes tier S	inside SEAL v2	name, size, hash on OFFER
File bytes tier M	private channel only — not a secret tier	entire file to anyone in channel
Exec argv / stdout	inside SEAL or DUMB-PSK	that an exec happened (INFO)
Jail contents	NT ACL + path jail	CAPA advertises jail root
TOFU pin	first write wins	a later attacker cannot replace the pin; they can only DoS new seals

Table 8 — Assets.

MUST NOT:

- Overwrite a pin on mismatch.
- Emit SEAL v1.
- Open Libera from CI or from pytest.
- Log AGPK/SEAL/DUMB bodies to stdout.
- Execute a job with no operator allowlist hit.
- Write complete files before hash check.
- Follow redirects, UNC paths, or execute downloaded files automatically after put.
- Disable TLS certificate verification. Default context as in irc_agent.py.

11. Build plan

Work is sequenced so each phase leaves main green. Do not start phase N+1 until phase N acceptance is checked. Estimated effort assumes one competent build agent with the existing tree checked out.

Phase 0 — Orient (0.5 day)

- 1 Clone the repo. Run `pip install -r requirements.txt` `pytest && pytest -q`. Confirm green.
- 2 Read SKILL.md, seal.py, irc_agent.py, protect.py, tests/*. Do not “clean up” style.
- 3 Create branch feat/moot-file-dumb from main.

Exit: `pytest -q` green on unmodified tree; branch exists.

Phase 1 — wire.py + CAPA (1 day)

- 1 Add scripts/wire.py with parsers for CAPA, MOOT, FILE, DUMB. Property: garbage in, None out.
- 2 Unit tests in tests/test_wire.py: happy path, extra fields, path-like ids, n too large, missing colon, nick regex fail, case of verbs (MOOT v1 open must fail; verbs are uppercase).
- 3 `irc_agent.handle_privmsg` calls `parse_capa_line` and INFO-logs a one-liner. No behaviour change for AGPK/SEAL.
- 4 `pytest -q` still includes every old test.

Exit: `test_wire.py` covers Table 3/6 verbs parse/reject; no Libera; old tests green.

Phase 2 — Moot (2 days)

- 1 scripts/moot.py CLI as §4.6. Writes outbox + moot json/txt.

- 2 `irc_agent.handle_moot` updates state. Chair/floor/roster rules as §4.3.
- 3 Skill leaflet `.grok/skills/agentic-moot/SKILL.md`.
- 4 `tests/test_moot.py`: state machine table tests with injected PRIVMSG strings; no sockets.
- 5 `install_skill.py` vendors the new skill + `moot.py`.

Exit: simulated 3-nick moot (chair, speaker, listener) in-process reaches CLOSE with a transcript file; listener that never held the floor emitted zero SAY lines.

Phase 3 — File sharing (2 days)

- 1 AIRC-FILE envelope encode/decode + basename jail.
- 2 `filexfer.py` offer/accept/refuse/status.
- 3 FileBag reassembly, hash check, complete/ write with `protect.py`.
- 4 `irc_agent.handle_file`.
- 5 Skill leaflet `agentic-file`.
- 6 `tests/test_filexfer.py`: tier S roundtrip using seal v2 helpers; tier M reassembly; hash mismatch does not write complete/; path traversal names rejected; `identity.json` offer refused.

Exit: 5 KiB random file tier S and 20 KiB tier M both land with matching sha256 in a tmp home.

Phase 4 — Dumb connector Python reference (2 days)

- 1 `seal.py` dumb-key + DUMB-PSK seal/open helpers (or `scripts/dumb_crypto.py`).
- 2 `dumb_agent.py`: connect loop copied in spirit from `irc_agent` (may import Client pieces if factored; factoring is allowed if tests stay green).
- 3 Job runner with jail, argv allowlist, timeout, single-flight.
- 4 `dumb_ctl.py` operator CLI.
- 5 Skill leaflet `agentic-dumb`.
- 6 `tests/test_dumb.py`: crypto roundtrip; jail escape rejected; allowlist miss dropped; exec of a fake argv runner via monkeypatch; put/get inside jail; busy rejection.

Exit: two temp homes, in-process (no Libera): `ctl` emits DUMB lines, dummy socket or direct `handle_privmsg` path runs `ping/sysinfo/exec-echo` and returns a result box.

Phase 5 — .NET 4.5 port (3 days)

- 1 `src/dumb_dotnet/` project, net45, AES-GCM via embedded library, SslStream TLS 1.2.
- 2 Behaviour tests in Python that spawn the exe only if DOTNET_DUMB_EXE is set; otherwise skip. Do not fail CI on machines without Framework 4.5.
- 3 docs in the skill: build with MSBuild, enable TLS 1.2, scheduled task XML skeleton.
- 4 `airc-dumb.cmd` wrapper.

Exit: on a modern Windows box with net45 targeting, the exe passes a local fake-IRC or recorded-handshake unit test. Full Libera join is a manual operator test, not CI.

Phase 6 — Integration polish (1 day)

- 1 README sections for the three features, still in field-kit voice. Keep the existing protocol block intact and add a new “Extensions” heading.
- 2 `install_skill.py` copies everything. Add a `--list` flag.
- 3 CI matrix remains ubuntu + py3.12 + `pytest -q`. Add no Libera service container.
- 4 Manual test script `tests/MANUAL.md` describing a 20-minute human session on a private channel (not run by CI).

Exit: README + skills + `pytest -q` green; `MANUAL.md` exists; this PDF’s acceptance list ticked.

11.1 Suggested commit series

```

1 test: capture baseline (optional empty)
2 feat: wire.py parsers + test_wire
3 feat: moot state, CLI, skill, test_moot
4 feat: file envelope, FileBag, CLI, skill, test_filexfer
5 feat: DUMB-PSK + dumb_agent.py + dumb_ctl.py + skill + test_dumb
6 feat: install_skill vendors new skills
7 feat: .NET 4.5 airc-dumb source and cmd wrapper
8 docs: README extensions + MANUAL.md

```

12. Test plan

Two layers: offline automated (CI) and a short manual Libera session (human). The build agent implements layer 1 fully and writes the layer 2 checklist. The build agent does not perform layer 2 against Libera from a cloud runner.

12.1 Automated — wire

ID	Case	Expect
W1	Each Table 3 verb, canonical spacing	parse success, fields populated
W2	moot verb lowercase / missing v1 / extra tokens	None
W3	FILE OFFER with path-like name ../x	None or refuse at offer time; never parse as valid name
W4	DUMB line with n=100	None (MAX_N 64)
W5	CAPA missing verbs=	None
W6	SEAL v2 still parsed only by seal.parse_seal_line	unchanged

Table 9 — Wire tests.

12.2 Automated — moot

ID	Case	Expect
M1	OPEN then JOIN×2 then ROSTER	roster length 3, state open
M2	SAY from non-floor in mode=floor	dropped, transcript unchanged
M3	FLOOR, SAY, YIELD *, chair CLOSE	state closed; further SAY dropped
M4	HANDOFF to nick not on roster	chair unchanged
M5	PART by floor holder	floor none
M6	Two OPENs with same id from different chairs	first wins; second INFO ignored
M7	seq not increasing for same nick	SAY dropped

Table 10 — Moot tests. Drive via `Client.handle_privmsg` with fake prefixes.

12.3 Automated — file

ID	Case	Expect
F1	Tier S 1 KiB known bytes	complete/ file sha256 match
F2	Tier M 20 KiB, shuffled chunk arrival	reassembled
F3	Hash mismatch DONE	no complete/ write; INFO fail
F4	offer of identity.json	CLI exit nonzero
F5	name with slash or space	rejected
F6	second OFFER same id	ignored
F7	ABORT mid-bag	no complete/; bag gone
F8	disk cap exceeded	ACCEPT not sent / refuse

Table 11 — File tests.

12.4 Automated — dumb

ID	Case	Expect
D1	PSK roundtrip ping	result ok
D2	from_nick not in --operators	no exec, no result on wire
D3	get path C:\agent-drop\..\Windows\win.ini	ok=false, error=jail
D4	put then get same bytes	sha256 match
D5	exec timeout	ok=false, error=timeout; child killed
D6	second exec while first runs	busy
D7	argv[0] not allowed	ok=false, error=bin
D8	meta characters without --allow-meta	rejected
D9	prefix != from_nick	drop
D10	wrong PSK	decrypt fail INFO, no exec

Table 12 — Dumb tests. Monkeypatch process runner; never spawn real cmd.exe that touches the CI disk outside tmp_path.

12.5 Automated — regressions

Re-run the full existing suite. Add a characterisation test that freeze-prints irc_lines_v2 for a fixed key material fixture if you can do so without committing a private key — prefer generate-in-test as today. protect.py tests stay platform-gated.

12.6 Manual — private Libera channel (human, 20 min)

- 1 Two modern laptops, two homes, one private channel, one human in HexChat watching pins.
- 2 Both announce AGPK. Human confirms first pins.
- 3 Open a floor moot, three SAY exchanges, close. Confirm transcripts match what HexChat saw.
- 4 Send a 3-line PowerShell snippet tier S. Receiver hash-checks.
- 5 On a lab Server 2012 (or a Windows VM with net45), start airc-dumb with PSK copied via RDP file copy. HexChat sees CAPA.
- 6 From laptop A: ping, sysinfo, put snippet, exec snippet, get results. Confirm jail refusal by asking for a path outside --allow-path.
- 7 Kill the connector with Ctrl+C; confirm reconnect INFO and a new CAPA.

- 8 Negative: from a third nick not on --operators, send a DUMB ping; connector stays quiet on channel and does not exec.
- Record the session only in the operator’s private notes. Do not commit irc.log. If SASL 903 appears, the skill may then say SASL worked for that nick; until then it stays unproven.

12.7 Pass/fail for the merge

Merge gate. pytest -q is green on Ubuntu / Python 3.12. No test opens a TCP connection to irc.libera.chat. New skills exist and install_skill.py copies them. README documents the three features in the existing voice. Jail tests exist and fail closed. The .NET tree builds on at least one documented command line, even if CI skips it.

13. Implementation defaults and constants

Name	Value	Where
FLOOD_S	0.8	irc_agent, dumb_agent, exe
CHUNK	300	seal, filexfer, dumb boxes
MAX_N (SEAL/DUMB)	64	seal.MAX_N
MAX_N_FILE	320	filexfer
BAG_TTL_S (SEAL/DUMB)	120	FragmentStore
FILE_TTL_S	1800	FileBag
FLOOR_IDLE_S	90	skill + optional chair hint only in v1
MOOT_ROSTER_MAX	16	wire / state
EXEC_TIMEOUT_DEFAULT	20	dumb
EXEC_TIMEOUT_MAX	60	dumb
RESULT_BYTES_MAX	8192	dumb boxed stdout
FILES_HOME_CAP	64 MiB	filexfer
TLS_HOST	irc.libera.chat	default
TLS_PORT	6697	default

Table 13 — Constants. Changing FLOOD_S requires a comment in README; Libera k-lines flooders.

14. Risks and mitigations

Risk	Impact	Mitigation
Stampede of agent replies	Channel flood, k-line, useless transcript	Floor mode default; skill forbids unsolicited SAY; flood 0.8s
Server 2012 TLS 1.2 disabled	Connector cannot join	Preflight in skill; document SchUseStrongCrypto; fail with INFO, do not downgrade TLS
PSK copied badly	Silent decrypt fail	dumb-key prints only a fingerprint (sha256 of key) to stderr; skill says compare fingerprints out of band
Tier M file on a “private” channel that is not	Data leak	Skill: tier M is clear; secrets are tier S or L
Connector as malware drop	Operator liability	Jail + no auto-exec of put files; allowlist; default deny operators
.NET AES-GCM quality	Crypto bug	Prefer a well-known portable library; Python is the reference; cross-test vectors in test_dumb.py
MAX_N_FILE still too small	Cannot move installers	Tier L drop path; do not raise MAX_N_FILE past 320 in v1
433 nick collision on 2012 box	Jobs addressed to original nick still work (match existing rule)	Reuse live_nick / original_nick logic exactly

Table 14 — Risks.

15. Handoff checklist for the build agent

- Do not reformat the existing scripts for taste. Match their style: no type-ignore soup, no extra deps, INFO-only stdout.
- Do not add asyncio, irc3, bottom, or any IRC library.
- Do not add a server. Do not add Docker required for unit tests.
- Do not claim SASL works.
- Do not generate images, dashboards, or a website.
- Do not open PRs that include identity.json, connector.key, irc.log, or real channel names beyond #ops / #your-channel placeholders.
- Do implement jail tests first if time is short — a connector without a jail is not shippable.
- If a choice is unspecified, pick the narrower capability and note it in README.
- When finished, run `pytest -q` and paste the summary in the PR body along with the phase that was reached.

16. Worked example (the session this kit is for)

Three nicks on private #ops: grok-box-a (chair, modern), claude-box (reviewer, modern), srv2012-box (dumb connector on the old file server). Human in HexChat.

```
grok-box-a      AGPK v1 <pk>
claude-box      AGPK v1 <pk>
srv2012-box    CAPA v1 dumb nick=srv2012-box verbs=ping,sysinfo,exec,get,put psk=1 agpk=0 jail=C:\agent-drop
grok-box-a      MOOT v1 OPEN a1a1a1a1a1a1a1a1 grok-box-a floor :patch the 2012 share
claude-box      MOOT v1 JOIN a1a1a1a1a1a1a1a1
grok-box-a      MOOT v1 FLOOR a1a1a1a1a1a1a1a1 grok-box-a
grok-box-a      MOOT v1 SAY a1a1a1a1a1a1a1a1 1 :@claude-box I will ping the box then put fix.ps1
grok-box-a      MOOT v1 YIELD a1a1a1a1a1a1a1a1 claude-box
claude-box      MOOT v1 SAY a1a1a1a1a1a1a1a1 1 :ack. I will review stdout only.
claude-box      MOOT v1 YIELD a1a1a1a1a1a1a1a1 grok-box-a
                  (off-transcript: dumb_ctl ping / put / exec; results sealed back to grok-box-a)
grok-box-a      MOOT v1 SAY a1a1a1a1a1a1a1a1 2 :@claude-box rc=0, hash ok, see SEAL id=bb..
grok-box-a      MOOT v1 CLOSE a1a1a1a1a1a1a1a1 :share patched
```

That is the whole product. If the build does not make this session possible on a private Libera channel with a Server 2012 jail, it is not done.

Appendix A — Existing function index to reuse

Symbol	File	Reuse
Client.handle_privmsg	irc_agent.py	Extend dispatch; keep AGPK/SEAL order
Client.say / outbox_loop	irc_agent.py	All new verbs go through outbox
Client.sasl_*	irc_agent.py	Do not touch
seal_bytes_v2 / open_bytes_v2	seal.py	Tier S files + AGPK-mode dumb jobs
irc_lines_v2 / FragmentStore	seal.py	Pattern for DUMB and FILE bags
tofu_pin / load_peers	seal.py	Unchanged
protect_path / write_secret_bytes	protect.py	homes, keys, complete files
NICK_RE / MSGID_RE	seal.py	Import in wire.py

Table 15 — Reuse map.

Appendix B — Out of scope until a later proposal

- Group SEAL / sender keys / any multi-recipient ciphertext.
- DCC SEND, HTTP file servers, BitTorrent.
- IRCv3 CHATHISTORY, so late joiners can reconstruct a moot.
- Automatic pin rotation, key-compromise protocol, signed AGPK.
- Running the dumb connector as SYSTEM with no jail.
- A GUI, a web client, or embedding The Lounge.
- Shipping a signed airc-dumb.exe from GitHub Actions in v1.
- Porting the operator skill runtime itself to Server 2012.

Appendix C — Acceptance one-pager

The work is accepted when all of the following are true:

- 1 pytest -q green, including new test_wire, test_moot, test_filexfer, test_dumb, and every test listed in §2.4.
- 2 Three new skills install via scripts/install_skill.py and mention the hard safety rules.
- 3 A 3-party in-process moot can be driven to CLOSE with floor discipline.
- 4 A 20 KiB file survives tier M reassembly with sha256 check.
- 5 A fake connector refuses jail escapes and unknown operators.
- 6 DUMB-PSK test vectors produced by Python open on the same blob the tests seal.
- 7 .NET project exists with a documented MSBuild command and TLS 1.2 noted as a Server 2012 preflight.
- 8 README still says field kit, not a platform, and still forbids unattended public channels.
- 9 No CI job resolves irc.libera.chat.

End of handoff. Implement phases 0–4 even if the .NET port slips; the Python connector plus this protocol is already enough for a modern box to drive another modern box, and the exe is the Server 2012 adapter on top of that frozen protocol.